MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6186W

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the electron tube described herein shall consist of this specification and the latest issue of MIL-E-1.

DESCRIPTION: Pentode, miniature, sharp cutoff

Outline	---	5-2 (EIA)
Base	---	E7-1
Envelope	---	T5-1/2
Cathode	---	Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7
Element	---	g1	k,g3, sd	h	h	a	g2	k,g3, sd

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Rk	Pg2	Pp	Ehk	Ik	TE	Alt
Unit:	V	V dc	V dc	V dc	Ohms	W	W	v	mA dc	°C	ft
Maximum:	6.9	330	---	250	---	0.55	2.5	100	---	+165	Note 2
Minimum:	5.7	---	---	---	---	---	---	---	Note 1	---	---
TEST CONDITIONS:	6.3	250	0	150	200	---	---	---	---	---	---

GENERAL:

Qualification - Required

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Method	Requirement or test	Notes	Conditions	AQL (percent defective)	Insp level or code	Symbol	Limits		Unit
							Min	Max	
	Quality conformance inspection, part 1								
1301	Heater current			0.4	II	If	275	325	mA
1336	Heater-cathode leakage			0.4	II	Ihk	---	10	µA dc
1266	Total grid current (1)	3,4		0.4	II	Ic1	0	-0.1	µA dc
1256	Electrode current (1) (anode)			0.4	II	Ib	5.2	8.8	mA dc
1256	Electrode current (screen)			0.4	II	Ic2	1.3	2.7	mA dc
1306	Transconductance (1)			0.4	II	Sm	4,000	6,000	µmhos
1201	Short and discontinuity detection			0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
1211	Insulation of electrodes			2.5	I	---	---	---	---
1256	Electrode current (2) (anode)		Rk = 0; Ec1 = -4.5 V dc	2.5	I	Ib	5	---	µA dc
1256	Electrode current (3) (anode)		Rk = 0; Ec1 = -10 V dc	2.5	I	Ib	---	100	µA dc
1306	Transconductance (2)		Ef = 5.7 V	2.5	I	ΔSm Ef	---	15	%
1266	Total grid current (2)	5	Ef = 7.0 V; Rg1 = 0.1 Meg	2.5	I	Ic1	0	-0.2	µA dc
2201	Noise and microphonics		Ef = 6.3 V ac; Ehk = 0; Ebb = Ecc2 = 300 V dc; Rg2 = 60,000 ohms; Rp = 10,000 ohms; RK = 200 ohms; Ecal = 50 mV ac	2.5	I	---	---	---	---
1331	Direct-interelectrode capacitance		No shield	6.5	Code E	Cgp	---	0.03	pF
						Cin	5.2	7.8	pF
						Cout	1.3	2.3	pF
1031	Low-frequency vibration		Rp = 2,000 ohms	6.5	Code H	Ep	---	100	mV ac
1031	Vibration-fatigue			6.5	Note 6	---	---	---	---

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Method	Requirement or test	Notes	Conditions	AQL (percent defective)	Insp level or code	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 2 -Continued								
1041	Shock		450 G; Ehk = 100 V dc; Rg1 = 0.1 Meg	6.5	Note 6	---	---	---	---
---	Vibration-fatigue and shock-test end points:								
1031	Low-frequency vibration		Rp = 2,000 ohms	---	---	Ep	---	450	mV ac
1336	Heater-cathode leakage			---	---	Ihk	---	30	µA dc
1306	Transconductance (1)			---	---	Sm	3,500	---	µmhos
1266	Total grid current (1)	4		---	---	Ic1	0	-0.2	µA dc
2126	Envelope strain			2.5	I	---	---	---	---
1121	Base strain	7		---	---	---	---	---	---
1105	Permanence of marking			---	---	---	---	---	---

Method	Requirement or test	Notes	Conditions	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 3						
1506	Heater-cycling life		Ef = 7.5 V; Ehk = 135 V dc; Ec2 = Eb = 0; 1 minute "on", 4 minutes "off"	---	---	---	---
---	Heater-cycling life-test end point:						
1336	Heater-cathode leakage			Ihk	---	30	µA dc
1516	Stability life		Ehk = 100 V dc; Rg1 = 1.0 Meg (min); TA = room	---	---	---	---
---	Stability life-test end point:						
1306	Change in transconductance (1) of individual tubes			ΔSm t	---	10	%

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Method	Requirement or test	Notes	Conditions	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 3 -Continued						
1501	Intermittent life	8,9	Ehk = 100 V dc; Rg1 = 1.0 Meg (min); TE = 165°C	---	---	---	---
---	Intermittent life-test end points (1,000 hours):						
1301	Heater current			If	275	325	mA
1336	Heater-cathode leakage			Ihk	---	10	µA dc
1266	Total grid current (1)	4		Ic1	0	-0.1	µA dc
1306	Transconductance (1)			Sm	3,500	6,000	µmhos
1306	Change in transconductance (1) of individual tubes			ΔSm t	---	25	%
1306	Transconductance (2)		Ef = 5.7 V	ΔSm Ef	---	20	%
1211	Insulation of electrodes			---	---	---	---

NOTES:

1. Difficulty may be encountered if tube is operated for long periods of time with very small values of cathode current.

2. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.

3. This test shall be performed at the conclusion of the holding period.

4. For an approved alternate test, the insertion of a 1.0 Meg resistor in the grid circuit shall not change the anode current by more than 1.0 mA dc.

5. Prior to this test, tubes shall be preheated a minimum of 5 minutes operating at the conditions indicated below. The 3-minute test shall not be permitted. Test at specified conditions within 3 seconds after preheating. Total grid current (2) shall be the last test performed on the sample selected for the total grid current (2) test.

Ef	Ec1	Ec2	Eb	Rk	Rg1
V	V dc	V dc	V dc	Ohms	Meg
7.0	0	150	250	200	0.25

6. This test shall be conducted on the initial lot and thereafter on a lot approximately every 12 months. When one lot has passed, the 12-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. MIL-STD-105, sample size code E, shall apply.

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NOTES: (Continued)

7. Acceptance sampling procedure shall be in accordance with method 1121, except that acceptance and rejection criteria shall be modified as follows:

   a. Accepted if not more than one defective for class A, B, or C defects, respectively, or if not more than a total of two defectives are found in the sample.

   b. Rejected if two or more defectives for class A, B, or C defects, respectively, or if a total of three or more defectives are found in the sample.

8. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (method 1226), will be satisfied if a TUT having bogey Ib (±5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.

9. The life-test sample shall consist of 20 tubes per lot and not more than 1 tube failure shall be permitted. In the event of rejection of the first sample, a second sample of 40 tubes shall be selected from the lot. Acceptance shall then be based on the combined first and second samples and shall not exceed a total of three tube failures.

10. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Air Force - 99
DLA - ES

User activities:
Army - ME
Navy - AS, CG, MC, OS, SH
Air Force - 11

Agent:
DLA - ES

Preparing activity:
Navy - EC

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